

My name is Muhanad Al-Khasawneh and I'm in a group with Muneer Al-Khasawneh and Mitch Koski. This semester for our senior design 1 we are going to work on a project to enhance a U.C. Core curriculum. The project is a 3D Printing Lab Management Platform modeled after 3D Print OS. It will be customized for the U.C. Core MVP the features include student submissions portal with in-browser slicer, UC email authentication with role based access, Job queueing and manager approval dashboard, airtable workflow integration for job tracking, printing compatibility, and toggleable cost visibility which will be for managers and admins only. For the roles I will be doing the frontend developing which i will be building the student facing website and slicer UI. Muneer will be doing the system engineering and he will develop manager dashboards, printers, monitoring, and maintenance tools. Mitch will be the lead integration engineer who's responsible for UC email authentication, airtable workflow, printer integration, and cost visibility toggle.

As a Computer Science student at the University of Cincinnati, I have completed a wide range of courses that gave me both the theoretical background and practical skills needed in the field. For example, CS2023: Programming Languages and Systems helped me strengthen my programming fundamentals and explore multiple paradigms, while CS4092: Database Design and Development allowed me to design and implement relational schemas using SQL and ER modeling. In CS4071: Network Security, I gained hands-on experience with encryption, session hijacking, and TCP/IP attack labs, which gave me a strong understanding of how to secure modern systems. I also expanded my knowledge of performance-focused computing through CS5153: Parallel Computing and CS6053: High Performance Computing, where I worked with CUDA, MPI, and OpenMP to analyze speedups and optimize algorithms. These courses not only developed my technical skills, but they also taught me important non-technical skills such as problem decomposition, teamwork, and technical communication through reports and group projects. Together, they provided a strong foundation that I can carry into my career, where designing efficient, secure, and user-friendly systems is critical.

My co-op experiences gave me the opportunity to apply these skills in real-world environments and build professional experience. I worked as a Project Coordinator Intern at Makino, where I used tools such as SAP, SharePoint, and Field Service Management to analyze sales data from 2022 onward. This role helped me strengthen my analytical and organizational skills as I calculated average job hours and costs while also learning to communicate findings clearly with engineers and managers. I also contributed to workflow improvement projects by building step-by-step job completion instructions for engineers, which required technical writing, attention to detail, and cross-team collaboration. Beyond Makino, I applied my computer science skills in

projects involving automation, data analysis, and dashboard creation using Excel and integrated systems. These experiences taught me how to balance technical development with professional communication, time management, and problem-solving under deadlines. I expect to apply these skills directly in my career by contributing not only as a developer, but also as a collaborator who can bridge the gap between technical solutions and organizational goals.

I am highly motivated to participate in this project because it blends my academic interests, technical skills, and the opportunity to make a meaningful contribution to the University of Cincinnati's Core curriculum. The idea of building a 3D Printing Lab Management Platform excites me because it directly benefits students and faculty by streamlining workflows that are currently fragmented or inefficient. This project also gives me the chance to strengthen my frontend development expertise while collaborating closely with my teammates, who bring complementary skills in system engineering and integration. The real-world impact of creating a student-facing website with features like an in-browser slicer and UC email authentication keeps me engaged because it feels like solving a problem with immediate value to the university community. Additionally, the complexity of the project pushes me to learn new tools, frameworks, and design strategies, which motivates me to stay innovative and flexible. I see this project not just as a class requirement, but as a stepping stone toward building professional-grade software solutions.

My preliminary approach to designing a solution is to break the project into clear, modular components so that each team member can work effectively while ensuring smooth integration at the end. On my side, I will focus first on creating wireframes and prototypes for the student portal and slicer UI, then iteratively refine them based on functionality and user experience goals. I expect the final product to include a fully operational submission platform, job queue management, and the slicer tool working seamlessly in-browser with compatibility across different printers. Success for me will mean that students and managers can interact with the system in a simple, intuitive way without confusion, while the backend integrates smoothly with the dashboards and Airtable workflows. I will evaluate my contributions by testing the frontend for responsiveness, reliability, and usability, as well as by gathering feedback from my teammates on how well the interface complements their components. I will know I have done a good job when the platform feels cohesive, performs reliably in real scenarios, and meets the goals we established at the beginning of the semester.